

2. (a) Explain what is meant by centripetal force and state its direction. [2]

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- (b) A small sphere of mass 30 g at the end of a light string rotates in a horizontal circle of radius 0.80 m and completes 10 revolutions in 15 s.

- (i) Show that the speed of the sphere is 3.35 m s^{-1} . [2]

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- (ii) Calculate the centripetal force on the sphere. [2]

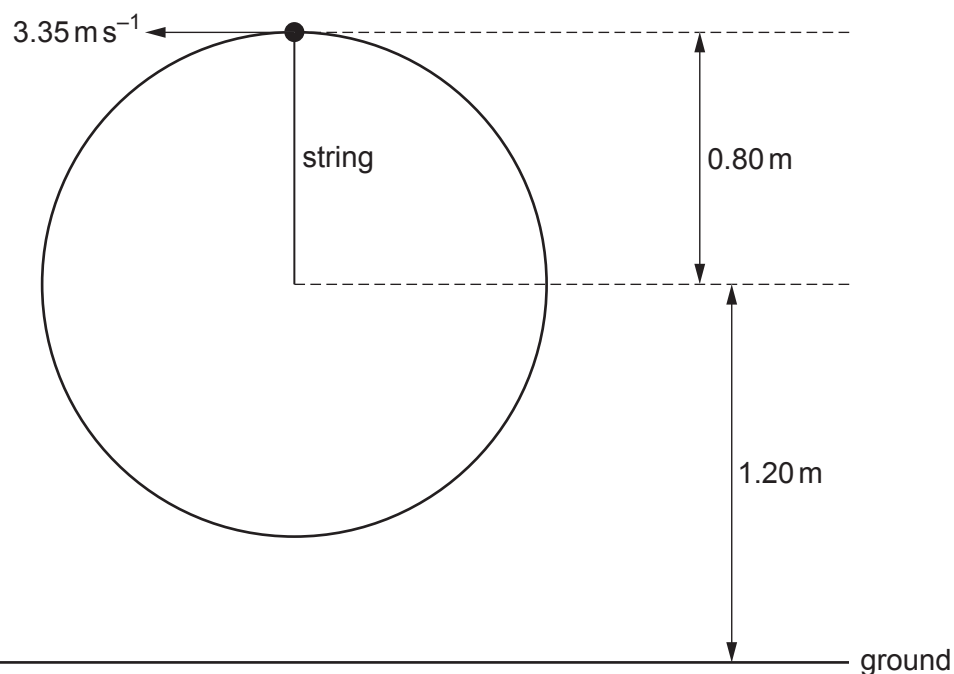
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- (c) The sphere now rotates in a **vertical** circle of the same radius, from a point that is 1.20 m above the ground. The speed at the top of the circle is 3.35 m s^{-1} .



Calculate the tension in the string at the **top of the circle**.

[2]

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- (d) (i) Calculate the speed of the sphere when it reaches the **bottom of the circle**. [3]

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- (ii) A student claims that if the string breaks when the sphere is at the **top of the circle** it will reach the ground at a horizontal distance of approximately 2 m away from the point of release. Investigate if her claim is correct, justifying your answer. [4]

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